

CMOS VLSI Design Project

Design and Analysis of a CMOS Full Adder

Using Cadence Virtuoso

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1 Introduction

The CMOS full adder is an important combinational building block used in arithmetic and digital systems. It adds three one-bit input signals and produces a Sum output and a Carry-out output.

In this project, a full adder was implemented and analyzed using the Cadence Virtuoso design environment. The circuit was constructed from hierarchical XOR, AND, and OR logic blocks. The design flow included schematic design, symbol generation, physical layout design, Design Rule Check (DRC), Layout Versus Schematic (LVS) verification, RC parasitic extraction (RCX), and post-layout simulation.

The post-layout simulation was used to examine the output waveforms and evaluate performance parameters such as average power consumption and propagation delay.

The Boolean expressions of a full adder are

$$S = A \oplus B \oplus C_{in} \quad (1)$$

and

$$C_{out} = AB + C_{in}(A \oplus B) \quad (2)$$

where A , B , and C_{in} are the input signals, S is the Sum output, and C_{out} is the Carry-out output.

2 Objectives

The main objectives of this project are:

- To implement a full adder using Cadence Virtuoso and reusable CMOS logic blocks.
- To create and verify the full-adder schematic.
- To generate a reusable symbol for the designed full adder.
- To design the physical layout of the full adder.
- To verify the physical layout using Design Rule Check (DRC).
- To verify the electrical equivalence between the schematic and layout using Layout Versus Schematic (LVS).
- To perform parasitic resistance and capacitance extraction using RCX.
- To perform post-layout transient simulation using the extracted circuit.
- To inspect the output waveforms of S and C_{out} and compare them with the expected full-adder truth table.
- To calculate the average power consumption and propagation delays of the full adder.

3 Theory of a Full Adder

A full adder is a combinational circuit that adds three one-bit input signals: A , B , and C_{in} . It produces two outputs, Sum (S) and Carry-out (C_{out}).

The Sum output is HIGH when an odd number of the three inputs are HIGH. Its Boolean expression is

$$S = A \oplus B \oplus C_{in}. \quad (3)$$

The Carry-out output is HIGH when at least two of the three input signals are HIGH. It can be expressed as

$$C_{out} = AB + AC_{in} + BC_{in}. \quad (4)$$

In the implementation used in this project, the carry expression is organized using the intermediate XOR term:

$$P = A \oplus B. \quad (5)$$

Therefore, the Sum output is

$$S = P \oplus C_{in} = A \oplus B \oplus C_{in}. \quad (6)$$

The Carry-out output is

$$C_{out} = AB + PC_{in} = AB + C_{in}(A \oplus B). \quad (7)$$

The truth table of the full adder is shown in Table 1.

Table 1: Truth Table of a CMOS Full Adder

A	B	C_{in}	S	C_{out}
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

The Sum output represents the least-significant bit of the addition, while the Carry-out output represents the carry generated by adding the three inputs.

4 Logic-Level Implementation

The full adder was assembled hierarchically using the XOR, AND, and OR logic blocks implemented in Cadence Virtuoso. The schematic contains two XOR blocks, two AND blocks, and one OR block.

Table 2: Logic Blocks Used in the Full-Adder Implementation

Logic Block	Quantity	Purpose
XOR2X1	2	Generate the Sum and intermediate XOR term
AND2X1	2	Generate the two carry terms
OR2X1	1	Combine the carry terms to produce C_{out}

First, the XOR block generates the intermediate signal

$$P = A \oplus B. \quad (8)$$

The second XOR block combines P with C_{in} to produce the Sum:

$$S = P \oplus C_{in}. \quad (9)$$

The first AND block generates AB , while the second AND block produces PC_{in} . The OR block combines these two terms to generate Carry-out:

$$C_{out} = AB + PC_{in}. \quad (10)$$

This hierarchical implementation reuses the logic blocks designed in the earlier circuit exercises.

5 Circuit Design

5.1 Schematic Design

The full-adder schematic was created using the Cadence Virtuoso Schematic Editor. The circuit has three inputs, A , B , and C_{in} , and two outputs, S and C_{out} .

The schematic connects two XOR blocks to generate the Sum output. Two AND blocks and one OR block form the carry-generation network.

The schematic of the designed full adder is shown in Figure 1.

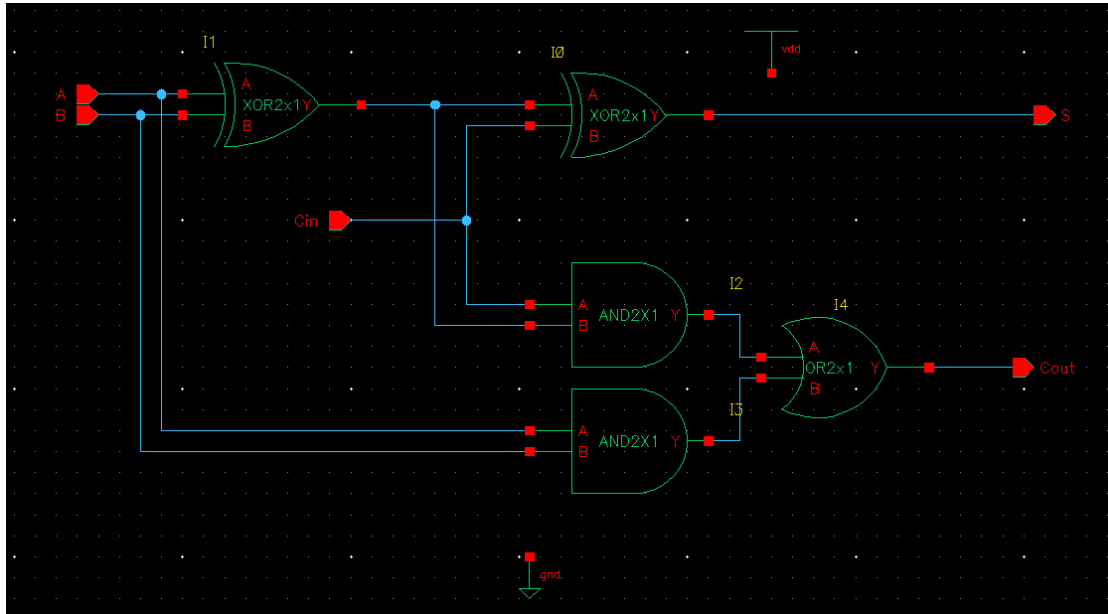


Figure 1: Schematic of the full adder implemented using XOR, AND, and OR logic blocks in Cadence Virtuoso.

The implemented Boolean functions are

$$S = A \oplus B \oplus C_{in} \quad (11)$$

and

$$C_{out} = AB + C_{in}(A \oplus B). \quad (12)$$

5.2 Full-Adder Symbol

After completing the schematic, a symbol was generated for the full adder. The symbol contains three input terminals labeled A , B , and C_{in} and two output terminals labeled S and C_{out} .

The generated full-adder symbol is shown in Figure 2.

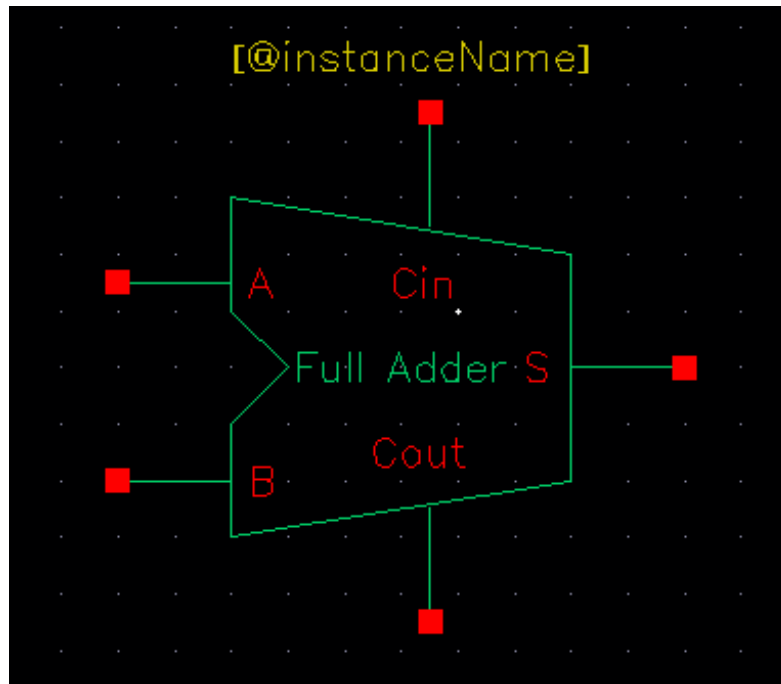


Figure 2: Generated symbol of the full adder.

The symbol allows the full adder to be used as a reusable hierarchical block in more complex digital circuits.

6 Physical Layout Design

After completing the schematic and symbol generation, the physical layout of the full adder was developed using the Cadence Virtuoso Layout Editor.

The layout represents the physical placement and interconnection of the devices used by the hierarchical logic blocks. It includes the required active or diffusion regions, polysilicon, well regions, metal layers, contacts, vias, and power and ground routing.

The completed physical layout is shown in Figure 3.

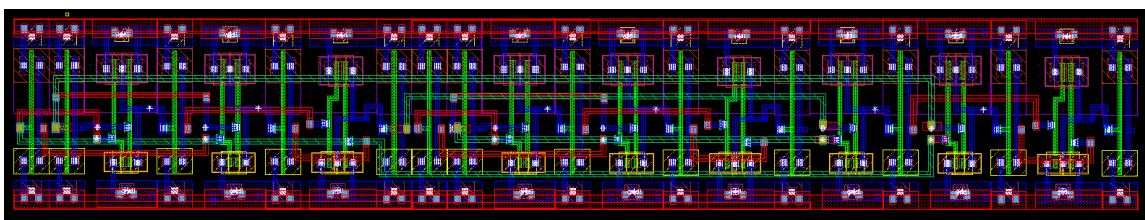


Figure 3: Physical layout of the full adder.

The width and height constraints above preserve the aspect ratio and keep the wide layout image within the page margins.

7 Design Rule Check (DRC)

Design Rule Check (DRC) is used to determine whether the physical layout satisfies the geometrical fabrication rules of the selected CMOS technology.

DRC checks constraints such as minimum metal width and spacing, polysilicon dimensions, diffusion spacing, contact enclosure, via enclosure, and well spacing.

The completed full-adder layout should be checked using the Cadence DRC verification flow. Record the actual result from the final Cadence run in the status box below.

DRC Result: PASSED

8 Layout Versus Schematic (LVS)

Layout Versus Schematic (LVS) verification compares the electrical netlist extracted from the physical layout with the netlist generated from the original schematic.

LVS checks the circuit connectivity and verifies that the layout corresponds to the intended full-adder implementation, including the input, output, power, and ground connections.

Record the actual LVS result from the final Cadence verification run in the status box below.

LVS Result: PASSED

9 RCX and Parasitic Extraction

After physical verification, RC parasitic extraction was performed using the RCX flow. Physical interconnections introduce parasitic resistance and capacitance that can affect circuit delay and power consumption.

The extracted full-adder view is shown in Figure 4.

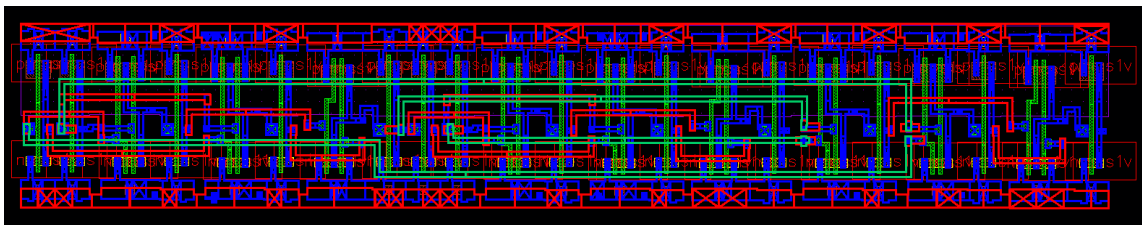


Figure 4: RCX/extracted view of the full adder.

The extracted view was used for the post-layout transient simulation and performance analysis described in the following sections.

10 Post-Layout Simulation

Post-layout transient simulation was performed using the extracted full-adder view in Cadence Virtuoso. This simulation includes the parasitic resistance and capacitance associated with the physical layout.

The following sections present the output waveform, average power calculation, and propagation-delay calculations.

11 Output Waveform

The output waveform was obtained from the transient simulation of the RC-extracted full adder. The waveform displays the input signals A , B , and C_{in} together with the Sum output S and Carry-out output C_{out} over the simulation interval.

The output waveforms are shown in Figure 5.

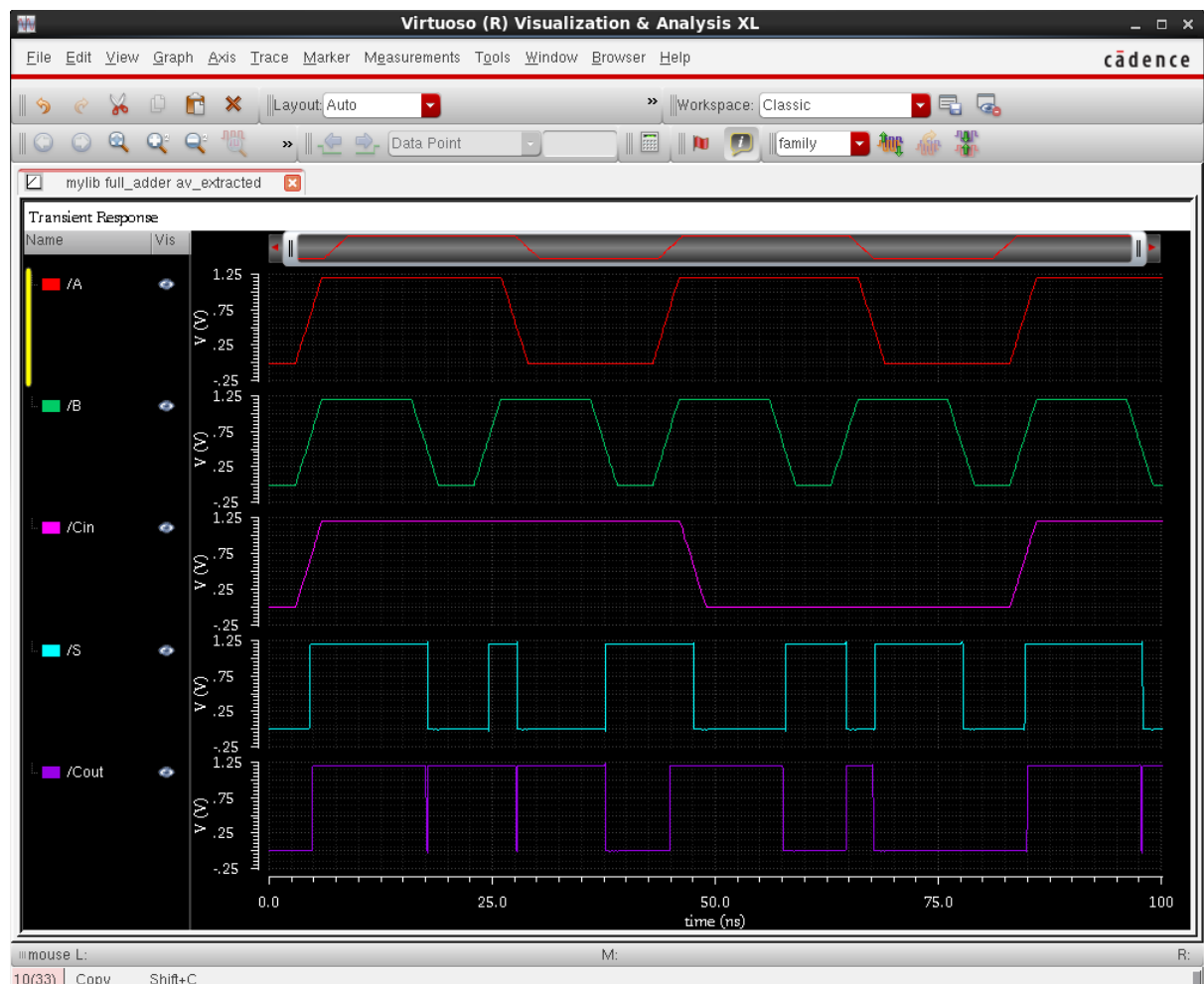


Figure 5: Post-layout transient waveforms of the full-adder inputs A , B , C_{in} and outputs S and C_{out} .

The Sum waveform follows

$$S = A \oplus B \oplus C_{in}, \quad (13)$$

while the Carry-out waveform follows

$$C_{out} = AB + C_{in}(A \oplus B). \quad (14)$$

The input and output traces can be compared with the truth table in Table 1 to verify the expected full-adder behavior for the simulated input combinations.

12 Average Power Calculation

The average power consumption of the full adder was calculated from the post-layout transient simulation using the Cadence Virtuoso Calculator.

The average power over a simulation interval T is expressed as

$$P_{avg} = \frac{1}{T} \int_0^T p(t) dt \quad (15)$$

where $p(t)$ represents the instantaneous power consumption.

The power calculation is shown in Figure 6.

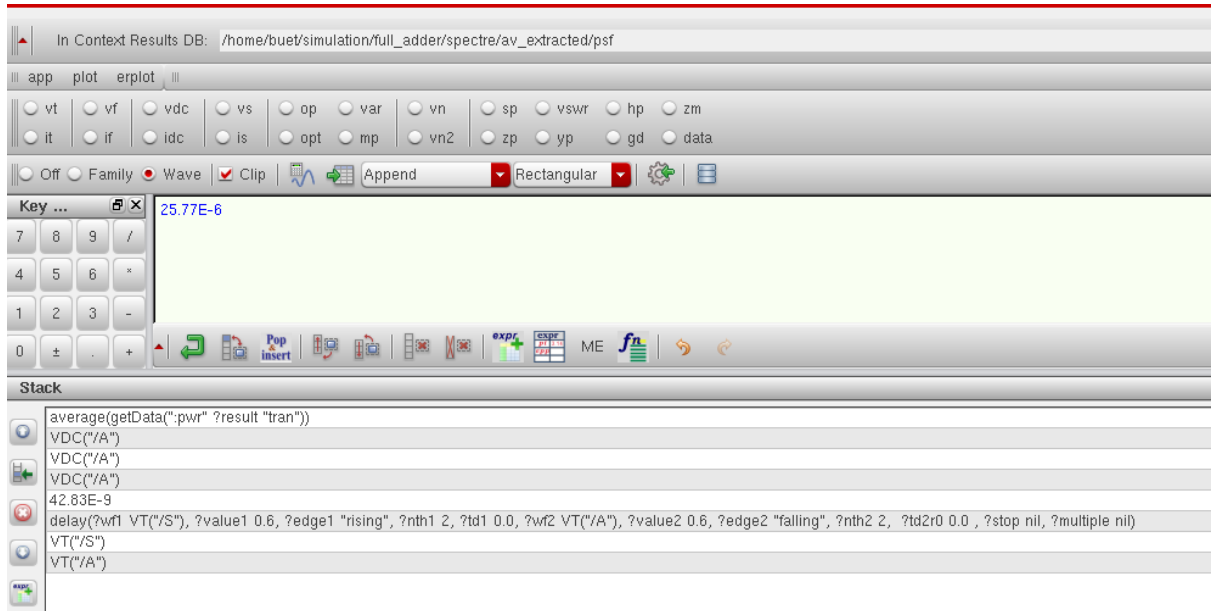


Figure 6: Average power calculation of the extracted full adder using the Cadence Virtuoso Calculator.

The displayed average power is

$$P_{avg} = 25.77 \times 10^{-6} \text{ W}. \quad (16)$$

Since

$$1 \mu\text{W} = 10^{-6} \text{ W}, \quad (17)$$

the average power consumption is

$$\boxed{P_{\text{avg}} = 25.77 \mu\text{W}}. \quad (18)$$

Therefore, the measured post-layout average power consumption of the full adder is approximately $25.77 \mu\text{W}$ under the applied simulation conditions.

13 Propagation Delay Calculation

Propagation delay represents the time required for an input transition to produce the corresponding transition at an output. Since the full adder has two outputs, separate selected input-to-output delays were measured for Sum and Carry-out.

A switching reference level of approximately 0.6 V was used in the Cadence Virtuoso Calculator. For a selected transition, the delay is calculated as

$$t_p = t_{\text{out}} - t_{\text{in}}. \quad (19)$$

13.1 Sum Output Delay

The selected delay calculation for the Sum output is shown in Figure 7.

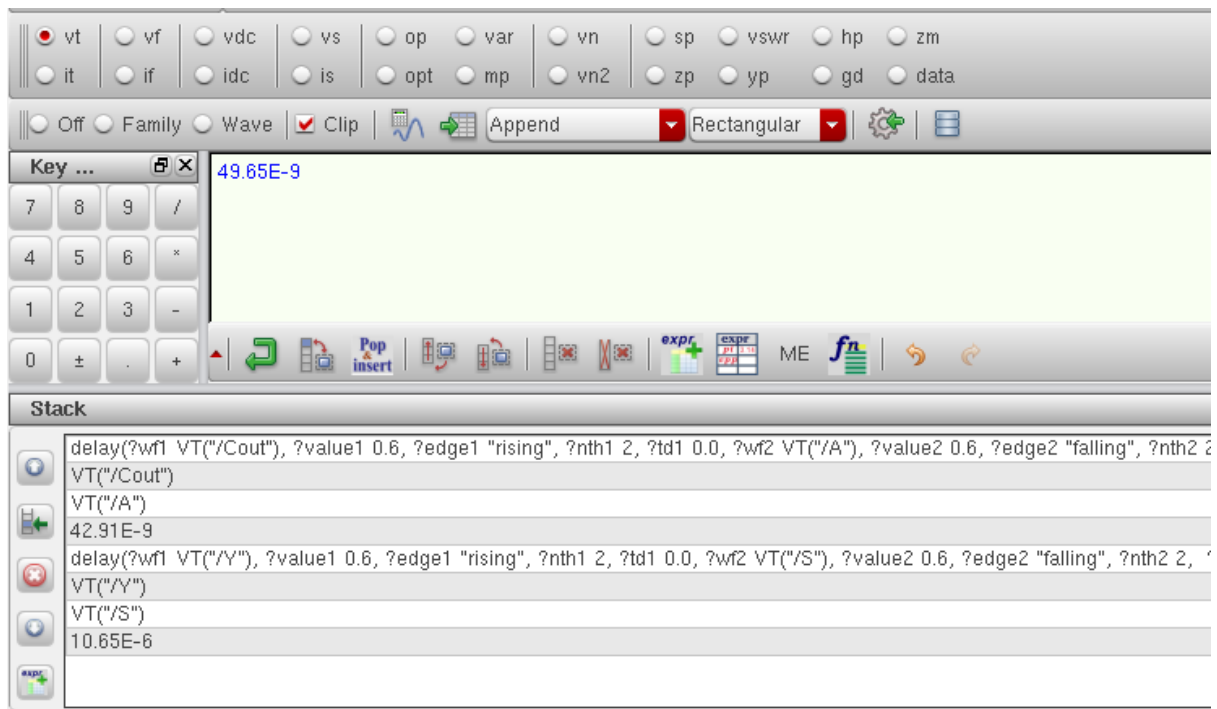


Figure 7: Cadence Calculator delay measurements for the full-adder outputs. The selected Sum-related readout is approximately 49.65 ns.

The selected Sum-output delay is

$$t_{p,S} = 49.65 \times 10^{-9} \text{ s.} \quad (20)$$

Since $1 \text{ ns} = 10^{-9} \text{ s}$, this corresponds to

$$t_{p,S} = 49.65 \text{ ns}. \quad (21)$$

13.2 Carry-Out Output Delay

The Carry-out delay calculation is shown in Figure 8.

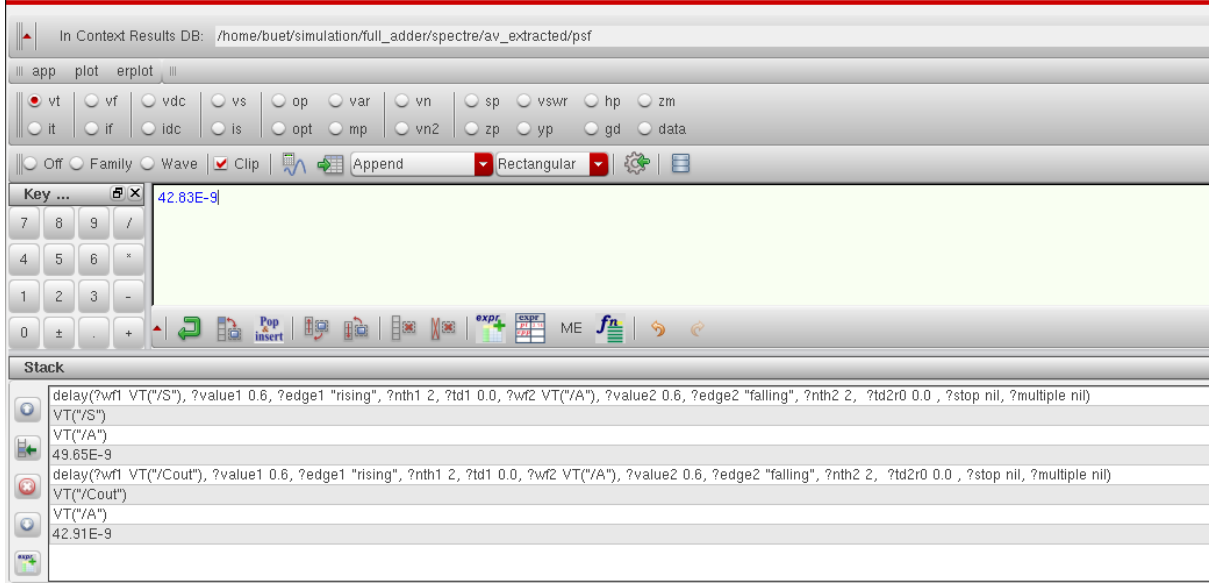


Figure 8: Cadence Calculator measurement for the full-adder Carry-out delay. The main readout shows approximately 42.83 ns.

The main Calculator readout shows a Carry-out delay of approximately

$$t_{p,Cout} = 42.83 \times 10^{-9} \text{ s.} \quad (22)$$

Therefore,

$$\boxed{t_{p,Cout} = 42.83 \text{ ns}.} \quad (23)$$

Measurement note: The Calculator expression stack in the provided screenshot also contains a value of 42.91×10^{-9} s for a Carry-out-related expression, while the main readout displays 42.83×10^{-9} s. Confirm the active expression and final measured value in Cadence before submitting the report; update the Carry-out value consistently if 42.91 ns is the intended result.

14 Verification and Performance Summary

The main design and post-layout simulation results of the full adder are summarized in Table 3.

Table 3: Summary of the Full-Adder Design Results

Parameter	Result
Logic Function (Sum)	$S = A \oplus B \oplus C_{in}$
Logic Function (Carry-out)	$C_{out} = AB + C_{in}(A \oplus B)$
Logic Blocks	2 XOR, 2 AND, 1 OR
Schematic Design	Completed
Symbol Generation	Completed
Physical Layout	Completed
DRC	Enter actual result
LVS	Enter actual result
RCX / Parasitic Extraction	Extracted view used for simulation
Post-layout Simulation	Completed
Average Power	25.77 μ W
Sum Output Delay	49.65 ns
Carry-out Delay	42.83 ns (verify readout)

15 Discussion

The CMOS full adder was implemented using Cadence Virtuoso and a hierarchical combination of XOR, AND, and OR logic blocks. The circuit accepts three one-bit inputs, A , B , and C_{in} , and produces two outputs, Sum (S) and Carry-out (C_{out}).

Two XOR blocks generate the Sum output and an intermediate XOR term. Two AND blocks generate the carry terms, and an OR block combines them to produce Carry-out. The resulting Boolean expressions are

$$S = A \oplus B \oplus C_{in} \quad (24)$$

and

$$C_{out} = AB + C_{in}(A \oplus B). \quad (25)$$

The physical layout was developed and an extracted view was used for post-layout transient simulation. The output waveform shows the three inputs and both outputs, allowing the simulated behavior to be compared with the expected truth table.

The average post-layout power consumption obtained from the Cadence Calculator was

$$P_{avg} = 25.77 \mu\text{W}. \quad (26)$$

The selected Sum-output delay was approximately

$$t_{p,S} = 49.65 \text{ ns}, \quad (27)$$

and the main Calculator readout showed a Carry-out delay of approximately

$$t_{p,Cout} = 42.83 \text{ ns}. \quad (28)$$

The Carry-out value should be confirmed against the saved Calculator expression stack, which also shows a 42.91 ns value. These delay values correspond to the selected transitions and simulation conditions rather than a universal delay for every possible input transition.

16 Conclusion

A CMOS full adder was implemented and analyzed using Cadence Virtuoso. The circuit combines two XOR blocks, two AND blocks, and one OR block to produce the Sum and Carry-out outputs from the three input signals.

The required Boolean functions are

$$S = A \oplus B \oplus C_{in} \quad (29)$$

and

$$C_{out} = AB + C_{in}(A \oplus B). \quad (30)$$

The project covered schematic design, symbol generation, physical layout design, physical verification, RC parasitic extraction, and post-layout transient simulation. A separate output-waveform section was included to document the simulated input and output signals.

The measured average power consumption was

$$\boxed{P_{\text{avg}} = 25.77 \text{ } \mu\text{W}}. \quad (31)$$

The selected Sum-output delay was approximately

$$\boxed{t_{p,S} = 49.65 \text{ ns}}, \quad (32)$$

while the main Calculator readout showed a Carry-out delay of approximately

$$\boxed{t_{p,Cout} = 42.83 \text{ ns}}. \quad (33)$$

The Carry-out delay should be confirmed against the saved Calculator expression before final submission. Overall, this project demonstrates the implementation, extraction,

waveform inspection, and performance analysis of a CMOS full adder using Cadence Virtuoso.

17 Software Used

The following software tools were used to complete the project:

- Cadence Virtuoso Schematic Editor
- Cadence Virtuoso Layout Editor
- Cadence Spectre Simulator
- Design Rule Check (DRC) Tool
- Layout Versus Schematic (LVS) Tool
- RCX / Parasitic Extraction Tool
- Cadence Virtuoso Visualization and Analysis Calculator